

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

The system of linear equations

$$3x - 2y - kz = 10$$

$$2x - 4y - 2z = 6$$

$$x + 2y - z = 5m$$

is inconsistent if :

(1) $k = 3, m = \frac{4}{5}$

(2) $k \neq 3, m \in \mathbb{R}$

(3) $k \neq 3, m \neq \frac{4}{5}$

(4) $k = 3, m \neq \frac{4}{5}$

Q2: 24 Feb (Shift 1) - Numerical

Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in \mathbb{R}$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to

Q3: 24 Feb (Shift 1) - Numerical

Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven, is

Q4: 24 Feb (Shift 2) - Single Correct

For the system of linear equations:

$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in \mathbb{R}$$

consider the following statements:

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if $k = -2$.

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(C) The system has unique solution if $k = 2$.

(D) The system has no-solution if $k = 2$.

(E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct?

(1) (B) and (E) only

(2) (C) and (D) only

(3) (A) and (D) only

(4) (A) and (E) only

Q5: 24 Feb (Shift 2) - Single Correct

Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2 B^2 - B^2 A^2) X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has :

(1) a unique solution

(2) exactly two solutions

(3) infinitely many solutions

(4) no solution

Q6: 25 Feb (Shift 1) - Numerical

If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2 + A)(I_2 - A)^{-1}$
 $= \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to

Q7: 25 Feb (Shift 1) - Numerical

If the system of equations

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$$kx + y + 2z = 1$$

$$3x - y - 2z = 2$$

$$-2x - 2y - 4z = 3$$

has infinitely many solutions, then k is equal to

Q8: 25 Feb (Shift 1) - Numerical

Let $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where x, y and z are real numbers such that $x + y + z > 0$ and $xyz = 2$. If $A^2 = I_3$,

then the value of $x^3 + y^3 + z^3$ is

Q9: 25 Feb (Shift 2) - Single Correct

If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is:

(1) 1

(2) 3

(3) 2

(4) 4

Q10: 25 Feb (Shift 2) - Single Correct

The following system of linear equations

$$3x + 3y + 2z = 9$$

$$3x + 2y + 2z = 9$$

$$x - y + 4z = 8$$

(1) does not have any solution

(2) has a unique solution

(3) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

(4) has infinitely many solutions

Q11: 25 Feb (Shift 2) - Single Correct

Let A be a 3×3 matrix with $\det(A) = 4$. Let R_i denote the i^{th} row of A . If a matrix B is obtained by performing the operation $R_2 \rightarrow 2R_2 + 5R_3$ on A , then $\det(B)$ is equal to:

- (1) 64
- (2) 16
- (3) 80
- (4) 128

Q12: 26 Feb (Shift 1) - Single Correct

The value of $\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$ is

- (1) -2
- (2) $(a+1)(a+2)(a+3)$
- (3) 0
- (4) $(a+2)(a+3)(a+4)$

Q13: 26 Feb (Shift 1) - Single Correct

Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is:

- (1) 6
- (2) 1
- (3) 4
- (4) 12

Q14: 26 Feb (Shift 2) - Single Correct

Consider the following system of equations :

$$x + 2y - 3z = a$$

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$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c$$

where a, b and c are real constants. Then the system of equations :

(1) has a unique solution when $5a = 2b + c$

(2) has infinite number of solutions when $5a = 2b + c$

(3) has no solution for all a, b and c

(4) has a unique solution for all a, b and c

Q15: 26 Feb (Shift 2) - Numerical

If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation $A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real

numbers α and β , then $\beta - \alpha$ is equal to

Answer Key**Q1** (4)**Q2** (17)**Q3** (540)**Q4** (3)**Q5** (3)**Q6** (13)**Q7** (21)**Q8** (7)**Q9** (1)**Q10** (2)**Q11** (1)**Q12** (1)**Q13** (3)**Q14** (2)**Q15** (4)

Q1 (4)

$$\Delta = \begin{vmatrix} 3 & -2 & -k \\ 1 & -4 & -2 \\ 1 & 2 & -1 \end{vmatrix} = 0$$

$$3(4 + 4) + 2(-2 + 2) - k(4 + 4) = 0$$

$$\Rightarrow k = 3$$

$$\Delta_x = \begin{vmatrix} 10 & -2 & -3 \\ 6 & -4 & -2 \\ 5m & 2 & -1 \end{vmatrix} \neq 0$$

$$10(4 + 4) + 2(-6 + 10m) - 3(12 + 20m) \neq 0$$

$$80 - 12 + 20m - 36 - 60m \neq 0$$

$$40m \neq 32 \Rightarrow m \neq \frac{4}{5}$$

$$\Delta_y = \begin{vmatrix} 3 & 10 & -3 \\ 2 & 6 & -2 \\ 1 & 5m & -1 \end{vmatrix} \neq 0$$

$$3(-6 + 10m) - 10(-2 + 2) - 3(10m - 6) \neq 0$$

$$-18 + 30m - 30m + 18 \neq 0 \Rightarrow 0$$

$$\Delta_z = \begin{vmatrix} 3 & -2 & 10 \\ 2 & -4 & 6 \\ 1 & 2 & 5m \end{vmatrix} \neq 0$$

$$3(-20m - 12) + 2(10m - 6) + 10(4 + 4) - 40m + 32 \neq 0$$

$$\Rightarrow m \neq \frac{4}{5}$$

Q2 (17)

$$\text{As } PQ = KI \Rightarrow Q = kP^{-1}I$$

$$\text{Now } Q = \frac{k}{|P|} (\text{adj} P) I$$

$$\Rightarrow Q = \frac{k}{(20+12\alpha)} \begin{bmatrix} - & - & - \\ - & - & (-3\alpha - 4) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ - & - & - \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\because q_{23} = \frac{-k}{8}$$

$$\Rightarrow \frac{k}{(20+12\alpha)} (-3\alpha - 4) = \frac{-k}{8}$$

$$\Rightarrow 2(3\alpha + 4) = 5 + 3\alpha$$

$$3\alpha = -3 \Rightarrow \alpha = -1$$

$$\text{Also } |Q| = \frac{k^3 |I|}{|P|}$$

Hints and Solutions

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$$\Rightarrow \frac{k^2}{2} = \frac{k^3}{(20+12\alpha)}$$

$$(20 + 12\alpha) = 2k \Rightarrow 8 = 2k$$

$$\Rightarrow k = 4$$

Q3 (540)

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$$a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 7$$

Case I : Seven (1's) and two (0's)

$${}^9C_2 = 36$$

Case II : One (2) and three (1's) and five (0's)

$$\frac{9!}{5!3!} = 504$$

$$\therefore \text{Total} = 540$$

Q4 (3)

$$x - 2y + 0 \cdot z = 1$$

$$x - y + kz = -2$$

$$0 \cdot x + ky + 4z = 6$$

$$\Delta = \begin{vmatrix} 1 & -2 & 0 \\ 1 & -1 & k \\ 0 & k & 4 \end{vmatrix} = 4 - k^2$$

For unique solution $4 - k^2 \neq 0$

$$k \neq \pm 2$$

For $k = 2$

$$x - 2y + 0 \cdot z = 1$$

$$x - y + 2z = -2$$

$$0 \cdot x + 2y + 4z = 6$$

$$\Delta_x = \begin{vmatrix} 1 & -2 & 0 \\ -2 & -1 & 2 \\ 6 & 2 & 4 \end{vmatrix} = (-8) + 2[-20]$$

Hints and Solutions

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$$\Delta x = -48 \neq 0$$

$$\text{For } k = 2 \quad \Delta x \neq 0$$

For $K = 2$; The system has no solution

Q5 (3)

$$A^T = A, B^T = -B$$

$$\text{Let } A^2 B^2 - B^2 A^2 = P$$

$$\begin{aligned} P^T &= (A^2 B^2 - B^2 A^2)^T = (A^2 B^2)^T - (B^2 A^2)^T \\ &= (B^2)^T (A^2)^T - (A^2)^T (B^2)^T \\ &= B^2 A^2 - A^2 B^2 \end{aligned}$$

$\Rightarrow P$ is skew-symmetric matrix

$$\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore ay + bz = 0$$

$$-ax + cz = 0$$

$$-bx - cy = 0$$

From equation 1,2,3

$$\Delta = 0 \& \Delta_1 = \Delta_2 = \Delta_3 = 0$$

$\therefore$ equation have infinite number of solution

Q6 (13)

$$A = \begin{bmatrix} 0 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 0 \end{bmatrix}$$

$$\Rightarrow I + A = \begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix}$$

$$\Rightarrow I - A = \begin{bmatrix} 1 & \tan \frac{\theta}{2} \\ -\tan \frac{\theta}{2} & 1 \end{bmatrix} \quad \left\{ \because |I - A| = \sec^2 \theta/2 \right\}$$

$$\Rightarrow (I - A)^{-1} = \frac{1}{\sec^2 \frac{\theta}{2}} \begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix}$$

Hints and Solutions

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$$\Rightarrow (I + A)(I - A)^{-1} = \frac{1}{\sec^2 \frac{\theta}{2}} \begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix}$$

$$= \frac{1}{\sec^2 \frac{\theta}{2}} \begin{bmatrix} 1 - \tan^2 \frac{\theta}{2} & -2 \tan \frac{\theta}{2} \\ 2 \tan \frac{\theta}{2} & 1 - \tan^2 \frac{\theta}{2} \end{bmatrix}$$

$$a = \frac{1 - \tan^2 \frac{\theta}{2}}{\sec^2 \frac{\theta}{2}}$$

$$b = \frac{2 \tan \frac{\theta}{2}}{\sec^2 \frac{\theta}{2}}$$

$$\therefore a^2 + b^2 = 1$$

Q7 (21)

$$D = 0$$

$$\Rightarrow \begin{vmatrix} k & 1 & 2 \\ 3 & -1 & -2 \\ -2 & -2 & -4 \end{vmatrix} = 0$$

$$\Rightarrow k(4 - 4) - 1(-12 - 4) + 2(-6 - 2)$$

$$\Rightarrow 16 - 16 = 0$$

Also. $D_1 = D_2 = D_3 = 0$

$$\Rightarrow D_2 = \begin{vmatrix} k & 1 & 2 \\ 3 & 2 & -2 \\ -2 & 3 & -4 \end{vmatrix} = 0$$

$$\Rightarrow k(-8 + 6) - 1(-12 - 4) + 2(9 + 4) = 0$$

$$\Rightarrow -2k + 16 + 26 = 0$$

$$\Rightarrow 2k = 42$$

$$\Rightarrow k = 21$$

Q8 (7)

$$A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix} \therefore |A| = (x^3 + y^3 + z^3 - 3xyz)$$

$$A^2 = I_3$$

$$|A^2| = 1$$

$$\therefore (x^3 + y^3 + z^3 - 3xyz)^2 = 1$$

Hints and Solutions

MathonGo

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz = 1$$

$$\Rightarrow x^3 + y^3 + z^3 = 6 + 1 = 7$$

Q9 (1)

$$\begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ -\alpha & \beta \end{bmatrix} = \begin{bmatrix} 1 + \alpha^2 & \alpha - \alpha\beta \\ \alpha - \alpha\beta & \alpha^2 + \beta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$1 + \alpha^2 = 1$$

$$\alpha^2 = 0$$

$$\alpha^2 + \beta^2 = 1$$

$$\beta^2 = 1$$

$$\alpha^4 = 0$$

$$\beta^4 = 1$$

$$\alpha^4 + \beta^4 = 1$$

Q10 (2)

$$\Delta = \begin{vmatrix} 2 & 3 & 2 \\ 3 & 2 & 2 \\ 1 & -1 & 4 \end{vmatrix} = -20 \neq 0 \quad \therefore \text{unique solution}$$

$$\Delta_x = \begin{vmatrix} 9 & 3 & 2 \\ 9 & 2 & 2 \\ 8 & -1 & 4 \end{vmatrix} = 0$$

$$\Delta_y = \begin{vmatrix} 2 & 9 & 2 \\ 3 & 9 & 2 \\ 1 & 8 & 4 \end{vmatrix} = -20$$

$$\begin{vmatrix} 2 & 3 & 9 \\ 3 & 2 & 9 \\ 1 & -1 & 8 \end{vmatrix} = -40$$

$$\therefore x = \frac{\Delta_x}{\Delta} = 0$$

$$y = \frac{\Delta_y}{\Delta} = 1$$

$$z = \frac{\Delta_z}{\Delta} = 2$$

Unique solution: (0,1,2)

Q11 (1)

Hints and Solutions

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$$A = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

$$2A = \begin{bmatrix} 2R_{11} & 2R_{12} & 2R_{13} \\ 2R_{21} & 2R_{22} & 2R_{23} \\ 2R_{31} & 2R_{32} & 2R_{33} \end{bmatrix}$$

$$B = \begin{bmatrix} R_2 \rightarrow 2R_2 + 5R_3 & & \\ 2R_{11} & 2R_{12} & 2R_{13} \\ 4R_{21} + 10R_{31} & 4R_{22} + 10R_{32} & 4R_{23} + 10R_{33} \\ 2R_{31} & 2R_{32} & 2R_{33} \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 5R_3$$

$$B = \begin{bmatrix} 2R_{11} & 2R_{12} & 2R_{13} \\ 4R_{21} & 4R_{22} & 4R_{23} \\ 2R_{31} & 2R_{32} & 2R_{33} \end{bmatrix}$$

$$|B| = \begin{vmatrix} 2R_{11} & 2R_{12} & 2R_{13} \\ 4R_{21} & 4R_{22} & 4R_{23} \\ 2R_{31} & 2R_{32} & 2R_{33} \end{vmatrix}$$

$$|B| = 2 \times 2 \times 4 \begin{vmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{vmatrix}$$

$$= 16 \times 4$$

$$= 64$$

Q12 (1)

$$C_1 \rightarrow c_1 - c_2, c_2 \rightarrow c_2 - c_3$$

$$= \begin{vmatrix} (a+2)a & a+1 & 1 \\ (a+3)(a+1) & a+2 & 1 \\ (a+4)(a+2) & a+3 & 1 \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \text{ \& } R_3 \rightarrow R_3 - R_1$$

$$= \begin{vmatrix} a^2 + 2a & a+1 & 1 \\ 2a+3 & 1 & 0 \\ 4a+8 & 2 & 0 \end{vmatrix}$$

$$= 6 - 8 = -2$$

Q13 (3)

Hints and Solutions

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$$\text{Let } A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$A^2 = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ab + bc \\ ab + bc & c^2 + b^2 \end{bmatrix}$$

$$= a^2 + 2b^2 + c^2 = 1$$

$$a = 1, b = 0, c = 0$$

$$a = 0, b = 0, c = 1$$

$$a = -1, b = 0, c = 0$$

$$c = -1, b = 0, a = 0$$

Q14 (2)

$$D = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 6 & -11 \\ 1 & -2 & 7 \end{vmatrix}$$

$$= 20 - 2(25) - 3(-10)$$

$$= 20 - 50 + 30 = 0$$

$$D_1 = \begin{vmatrix} a & 2 & -3 \\ b & 6 & -11 \\ c & -2 & 7 \end{vmatrix}$$

$$= 20a - 2(7b + 11c) - 3(-2b - 6c)$$

$$= 20a - 14b - 22c + 6b + 18c$$

$$= 20a - 8b - 4c$$

$$= 4(5a - 2b - c)$$

$$D_2 = \begin{vmatrix} 1 & a & -3 \\ 2 & b & -11 \\ 1 & c & 7 \end{vmatrix}$$

$$= 7b + 11c - a(25) - 3(2c - b)$$

$$= 7b + 11c - 25a - 6c + 3b$$

$$= -25a + 10b + 5c$$

$$= -5(5a - 2b - c)$$

$$D_3 = \begin{vmatrix} 1 & 2 & a \\ 2 & 6 & b \\ 1 & -2 & c \end{vmatrix}$$

$$= 6c + 2b - 2(2c - b) - 10a$$

Hints and Solutions

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$$= -10a + 4b + 2c$$

$$= -2(5a - 2b - c)$$

for infinite solution

$$D = D_1 = D_2 = D_3 = 0$$

$$\Rightarrow 5a = 2b + c$$

Q15 (4)

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{19} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{19} & 0 \\ 3 & 0 & -1 \end{bmatrix}, A^{20} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{20} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{L.H.S} = A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 + \alpha + \beta & 0 & 0 \\ 0 & 2^{20} + \alpha 2^{19} + 2\beta & 0 \\ 3\alpha + 3\beta & 0 & 1 - \alpha - \beta \end{bmatrix}$$

$$\text{R.H.S} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \alpha + \beta = 0 \text{ and } 2^{20} + \alpha 2^{19} + 2\beta = 4$$

$$\Rightarrow 2^{20} + \alpha (2^{19} - 2) = 4$$

$$\Rightarrow \alpha = \frac{4 - 2^{20}}{2^{19} - 2} = -2$$

$$\therefore \beta - \alpha = 4$$

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